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Omkar Bhilare

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Summary — I combine research, industry, teaching experience, and hands-on projects to address complex problems in hardware architectural design, verification, CAD optimization, and FPGA applications for scalable digital systems.

Education

University of Toronto (UofT), Toronto, Canada — ***Advisor:*** *Prof. Jason Anderson*
PhD candidate in ECE [**GPA**: 3.86/4] — *September 2023 - Present*

→ **Relevant Coursework: ECE552** (Computer Arch.) || **ECE1718** (Advanced Computer Arch.) || **ECE1756** (Reconfigurable Computing & FPGA Arch.) || **ECE1755** (Parallel Computer Arch. & Prog.) || **ECE1387** (CAD for Digital Circuits Synthesis) || **ECE1777** (Circuit Simulation) || **ECE1528** (IoT Protocols & Applications)

Bachelor's in Engineering (EE), VJTI, Mumbai, India [**GPA**: 9.13/10] — *July 2019 - May 2022*

Publications

[1] R. Ide, T. Ahmed, B. Adhi, S. Kawai, **O. Bhilare**, J.H. Anderson, "Hardware Sharing for Area Optimization of Floating-Point ALUs," *HEART '26*
[2] **O. Bhilare**, O. Ragheb, B. Adhi, K. Sano, J.H. Anderson, T. Ueno, "Predication in Elastic CGRAs," *CGRA4HPCA, IPDPS '26*
[3] H. Wei, **O. Bhilare**, H. Waqar, J.H. Anderson, "CAD Techniques for NoC-Connected Multi-CGRA Systems," *HEART '24*
[4] **O. Bhilare**, and others, "DEEPFAKE CLI: Accelerated Deepfake Detection Using FPGAs," *PDCAT 2022*

Invited Talks

→ "Adaptive Data Compression on CGRAs," *2nd Research Meeting on Adaptive Bandwidth Compression Hardware (ABC-HW '26)*, Tohoku University, Sendai, Japan (remote) — *March 2026*

Research Experience

Processor Research Team, RIKEN-CCS, Kobe, Japan — Dr. Kentaro Sano, Dr. Tomohiro Ueno
Research Intern — *January 2026 - March 2026*

→ Extended an elastic CGRA architecture and its CAD flow to support predicated execution, enabling acceleration of compute-intensive loops containing conditional control-flow

→ Re-purposed existing crossbar components to add predication support, minimizing area overhead to just 6–20% across partial and full predication schemes while leaving the critical-path delay unchanged

→ Applying predication support to implement adaptive compression/decompression on CGRAs *(Work-in-progress)*

Processor Architecture Laboratory (LAP), EPFL, Switzerland — Prof. Paolo Ienne, Mr. Andrea Guerrieri
Summer@EPFL — *June 2022 - August 2022*

→ Developed a framework to enable Dynamatic (a dynamic HLS compiler) to access a multi-cycle memory system via a custom AXI interconnect, integrated with its complex out-of-order load-store queues (LSQs)

→ Enhanced memory accessing capabilities of Dynamatic with only a 2% increase in LUT and a 4% in FF usage

Shakti Lab, RISE Group, IIT Madras, India — Prof. V. Kamakoti, Mrs. Lavanya J
Research Verification Intern [REPORT] — *March 2021 - July 2021*

→ Designed an FPGA-framework for verifying SHAKTI, India's first indigenous RISC-V processor family

→ Enhanced verification speed to 500k RISC-V instructions in 64 seconds on FPGA— orders of magnitude faster than conventional simulation

CoE in Complex Dynamical Systems, VJTI — Prof. Faruk Kazi
Research Associate [Bachelor's thesis] — *June 2021 - May 2022*

→ Trained, quantized, and compiled various AI models for deployment on Xilinx Zynq and Versal FPGA platforms

→ Achieved a **120% improvement** in the Deepfake Detection model's inference speed on the VCK5000 compared to the state-of-the-art Nvidia Tesla T4 GPU

Projects

Determining and bridging area gap between ASIC and CGRA *Aug 2024 - Present*

→ Analyzing programmability costs in modern CGRA architectures by comparing Power, Performance, and Area results from diverse benchmark suites run on both ASIC and CGRA platforms

→ Conducting architectural exploration to identify trade-offs between programmability and performance in CGRAs

→ Bridging the power gap in CGRA architectures by leveraging fracturable processing elements and analyzing their impact on Performance and Area *(Work-in-progress)*

NoC-Connected Multi-CGRA Systems *Jan 2024 - May 2024*

→ Developed a partitioning-based CAD flow for NoC-connected large CGRAs, reducing run time by 33× compared to traditional CAD methods

→ Integrated and customized an open-source min-cut partitioner, TritonPart, to fit specific CGRA CAD requirements

Skills

Language: Verilog, VHDL, C, C++, Python **Systems:** Git, Linux
EDA Tools: Quartus Prime, Xilinx Vivado, Microchip Libero, Yosys, Synopsys Design Compiler, Cadence Innovus

Work Experience

AMD **Bangalore, India**
Co-Op + Silicon Design Engineer I *Dec 2021 - July 2023*

→ Contributed to SoC-level verification of Design for Debug (DFD) IPs in 3 AMD processors, aiding in successful tapeouts

→ Verified debug infrastructure in IPs, including USB, memory, and CPU cores, etc., using constrained-random test cases, ensuring functional correctness

→ Developed and ported a verification testsuite for DFD IPs across AMD processors, ensuring maximum test coverage and robustness

Google Summer of Code 2021, BeagleBoard Organization **Remote**
Open-Source Developer [REPORT] *June 2021 - Aug 2021*

→ Designed and tested gateware for BeagleWire, a Lattice iCE40 FPGA cape for the BeagleBone Black single-board computer

→ Established seamless communication between the ARM processor and the FPGA using General Purpose Memory Controller (GPMC) and Wishbone protocols

→ Integrated BeagleWire with SDRAM by leveraging the LiteDRAM core, enabling memory initialization through SERV, a lightweight, bit-serial RISC-V CPU

Teaching Experience

Teaching Assistant for **ECE1387** (CAD for Digital Circuit Synthesis) [F'24] || **ECE532** (Digital Systems Design) [W'24, S'24, W'25] || **ECE241** (Digital Systems) [F'23, F'24, F'25] || **CSC258** (Computer Organization) [F'24] — mentored semester-long FPGA design projects, supervised digital-circuit labs, graded advanced CAD algorithm assignments, and integrated a new Microchip FPGA discovery kit into the ECE532 curriculum.

Awards AND Recognition

→ **Adaptive Computing Challenge 2021 by AMD-Xilinx** (3rd place, $3000 prize) – Top 3 out of 165 entries

→ ***Summer@EPFL*** Scholar: Three-month summer fellowship to conduct research at EPFL

→ Selected for the prestigious ***Google Summer of Code*** program, aimed at encouraging new contributors to open-source development.